

REVIEW ARTICLE

GLP-1 Receptor Agonists

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SUMMARY

Glucagon-like peptide-1 (GLP-1) receptor agonists are incretin analogues that promote glucose-mediated insulin release and are used to treat type 2 diabetes mellitus and obesity. GLP-1 receptor agonists and GLP-1 and glucose-dependent insulinotropic peptide agonists have several mechanisms of action, including reduction of gastric emptying, inhibition of glucagon secretion, beneficial changes in the intestinal microbiome, and direct effects on hypothalamic nuclei to enhance satiety (which promotes weight loss). Beyond the impressive effects of GLP-1 receptor agonists on blood glucose levels and body weight, large-scale randomized, controlled trials have shown that GLP-1 receptor agonists reduce cardiovascular risk and slow progression to renal failure in persons at high risk and those with type 2 diabetes. Adverse side effects from GLP-1 receptor agonists are mostly gastrointestinal but may also include loss of muscle and bone mass. Questions remain about long-term adherence, weight regain after discontinuation of treatment, and the functional implications of the loss of muscle and bone mass. Recent and ongoing targeted studies suggest the possibility of additional uses for GLP-1 receptor agonists.

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CURRENTLY, THERE ARE MORE THAN 1 BILLION PEOPLE WORLDWIDE with obesity, many of whom also have type 2 diabetes mellitus or prediabetes.¹ The recent emergence of novel therapies that drive weight loss and improve glucose control have generated hope and expectation that these conditions ultimately will be controlled. Yet, with the advent of these new agents comes the clinical dilemma of how to use them in a safe, cost-effective manner.² This review describes a new generation of treatment for persons with obesity with or without diabetes or prediabetes.

DISCOVERY AND DRUG DEVELOPMENT

The discovery of insulinotropic peptides parallels the early history of insulin. In the late 19th century, laboratory work involving animal models showed that removal of the pancreas led to high blood glucose levels, wasting, and, ultimately, death. It took 30 years before Banting and Best isolated and purified insulin and successfully used it to treat a person with diabetes.³ Shortly thereafter, a British group discovered that molecules isolated from the intestine could lower blood glucose levels and named them incretins (now called incretins).⁴ Subsequently, the remarkable efficacy of insulin in the treatment of persons with high glucose levels overshadowed work on incretins until seminal studies by Habener, Drucker, Holst, Bloom, and others showed that incretins were potent insulinotropic molecules.⁵⁻¹² Collectively, these studies identified and characterized glucagon-like peptide-1 (GLP-1) in intestinal L cells as a peptide product cleaved from glucagon (amino acids 7 to 37). In 1990, John Eng isolated a native peptide from the venom of the Gila monster lizard, exendin-4, which is structurally similar to GLP-1 and competes with GLP-1 for its receptor, stimulates insulin secretion, and has a longer half-life.¹²⁻¹⁴ The glucose-lowering effects of

exendin-4 were established in clinical trials.¹⁵ In 2005, exenatide became the first incretin analogue approved for the treatment of type 2 diabetes.¹⁶⁻¹⁸ Concomitantly, various pharmaceutical companies focused their efforts on generating synthetic GLP-receptor analogues to treat type 2 diabetes by extending the half-life of GLP-1 through modifications to amino acids and resisting cleavage by dipeptidyl peptidase-4 *in vivo*.^{19,20}

Glucose-dependent insulinotropic peptide (GIP) was the first incretin isolated.^{21,22} GIP is synthesized in intestinal K cells and enhances glucose-mediated insulin release. Although GIP administration led to only modest changes in blood glucose levels in patients with type 2 diabetes, it exerts a prosurvival effect on beta cells in pancreatic islets. When GIP was combined with synthetic GLP-1 receptor agonists, additive effects on both glucose levels and body weight occurred. GIP was also found to have very modest insulin-sensitizing effects on adipocytes. Together, these efforts facilitated a new era of GLP-1-based therapeutics that now include double and triple agonists, several of which have been approved by international regulatory agencies, with more in the pipeline for the treatment of type 2 diabetes and obesity (Table 1).

SECRETION AND BINDING OF GLP-1 AND GIP

The incretin family is generally considered to include GIP, GLP-1, and glucagon.²¹ All are produced and secreted by discrete intestinal cells in response to meals but are also synthesized elsewhere, particularly in the brain stem. GLP-1 and GIP are secreted in response to gut glucose levels, whereas intravenously administered glucose produces negligible changes in incretin release (Fig. 1). These gut peptides are released in response to ingested proteins and lipid products when glucose levels are elevated, enhancing insulin release at critical times.²³ The half-lives of circulating GLP-1 and GIP are quite short owing to gut degradation, systemic dipeptidyl peptidase-4 enzymatic activity, and rapid renal clearance.

GLP-1, GIP, and glucagon each has its own receptor distributed within various tissues throughout the body. Because the primary mode of action is to stimulate insulin secretion in a glucose-dependent manner, pancreatic islet beta cells have the highest concentration of receptors for these peptides.²¹ GLP-1 is activated by binding to a G-protein-coupled receptor, which triggers the release of intracellular cyclic AMP (cAMP) and activation of adenylyl cyclase.²⁴ This event drives increased Ca^{2+} uptake through voltage-dependent calcium channels as well as stimulation of Ca^{2+} release from intracellular stores that involve Epac2 (exchange protein directly activated by cAMP 2) activation in the beta cell. GIP binds to its own G-protein-coupled receptor and activates identical pathways, although with a greater dependency on K^{+} ATP channels than GLP-1 in islet cells.^{21,24} This rich set of pathways and interactions has provided approachable targets for control of both hyperglycemia and excessive weight.

GLUCOSE-LOWERING PROPERTIES OF GLP-1 AND OTHER INCRETIN RECEPTOR AGONISTS

GLP-1 and GIP promote insulin secretion directly but also inhibit glucagon and stimulate beta-cell prosurvival and proliferation mechanisms to maintain islet cell mass.^{21,24} GLP-1 also slows gastric emptying and reduces gastric motility, both of which can lower plasma glucose

Table 1. Side Effects Associated with GLP-1 Receptor Agonists.*

Adverse events
Gastrointestinal — nausea, vomiting, abdominal pain, diarrhea, constipation, bloating, belching, flatulence
Glycemic — hypoglycemia in persons with diabetes mellitus
General well-being — fatigue, dizziness
Central nervous system — headache, mood changes
Drug interactions
Serious adverse events
Allergy — anaphylaxis
Pancreatic — acute pancreatitis
Gallbladder — acute cholelithiasis
Kidney — acute kidney injury
Mood — mood changes, suicidality
Tumorigenesis — risk of thyroid C-cell tumors (only in animal models to date)
Ophthalmologic — risk of worsening diabetic retinopathy

* The list of adverse events shown is not exhaustive; the adverse events and side effects shown here are ones that are reported most commonly. Serious adverse events are rare. GLP-1 denotes glucagon-like peptide 1.

levels. There is some evidence that GLP-1 receptor agonists could modulate gut microbiota, thereby improving glucose tolerance, although more studies are needed.²⁵ GLP-1 receptor agonists have minimal effects on insulin sensitivity in muscle or adipose tissue but have other indirect effects on glucose tolerance through changes in fat metabolism. On the other hand, GIP can activate AKT (protein kinase B), which promotes the translocation of glucose transporter 4 to the membrane, enhancing glucose uptake in adipocytes.²⁶ GIP secretion can promote lipid storage, thereby reducing circulating free fatty acids by increasing new lipogenesis and stimulating adipocyte lipoprotein lipase.^{27,28} These properties enhance the effectiveness of dual therapy to promote glucose tolerance with GLP-1 receptor agonists and GIP receptor agonists.

In contrast to GIP and GLP-1, glucagon itself, a counterregulatory hormone that raises blood glucose levels by stimulating glycogenolysis and gluconeogenesis in the liver, also increases satiety owing to its direct effect in the ventromedial hypothalamus.²⁶ It is noteworthy that glucagon or its agonists, which act through the glucagon receptor, are potentially useful adjuncts in fixed combinatorial approaches, particularly because glucagon can also reduce intrahepatic fat infiltration and promote lipolysis. Taken together, the GLP-1 receptor agonists with or without GIP agonists have profound effects on blood glucose dynamics through direct stimulation of insulin release and indirect actions on fat metabolism, gastric emptying, and gut motility.

GLP-1 RECEPTOR AGONISTS AND THE CENTRAL NERVOUS SYSTEM

Regulation of hypothalamic neural centers for both glucose control and weight maintenance are characteristic features of the GLP-1 receptor agonists. GLP-1 receptors are abundantly present in the arcuate nucleus, which contains agouti-related protein and neuropeptide Y cells, as well as in hypothalamic areas where GLP-1 is also produced, although GLP-1 is thought to be generated principally in the brain stem. Receptors for the incretins control body weight and food intake separate from control of blood glucose. Intracerebroventricular injection of GLP-1 directly into the paraventricular nucleus leads to satiety,

but because pro-opiomelanocortin nuclei also express GLP-1 receptors, there is an anorexic effect.²⁹ The arcuate nucleus (but not the paraventricular nucleus), with its abundant GLP-1 receptors, regulates glucose levels by mediating hepatic glucose output. In addition, there is evidence that substrate utilization is altered by GLP-1 in both the paraventricular and arcuate nucleus, with a shift toward lipid oxidation and lower respiratory exchange ratios.³⁰

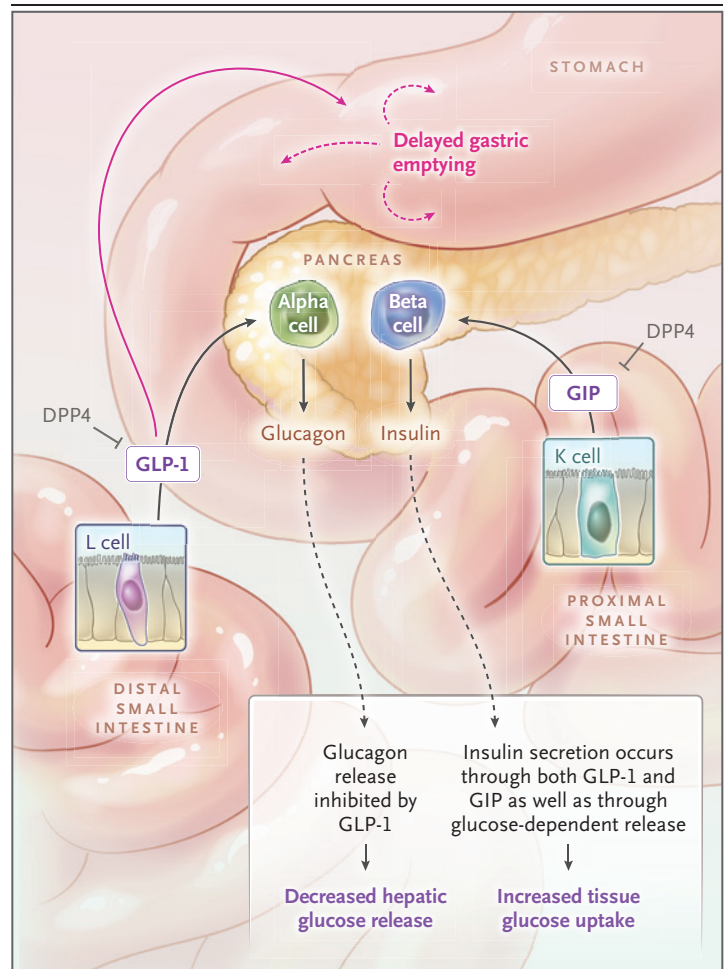


Figure 1. Physiologic Effects of the Incretin Family of Drugs.

The three major insulinotropic peptides (glucagon-like peptide 1 [GLP-1], glucose-dependent insulinotropic peptide [GIP], and glucagon) are synthesized by several tissues, although the pancreas and the small intestine are major sources of the circulating forms. Synthesis begins with the 160–amino acid glucagon precursor, proglucagon (PG), which is differentially processed in the pancreas but not in the gut and brain. Proglucagon is cleaved to glucagon in the pancreas, and in the gut, four fragments having important functions are produced.

EFFECT ON GLYCATED HEMOGLOBIN LEVELS

Randomized, placebo-controlled trials of the GLP-1 receptor agonists have provided strong evidence to support their use — in fact, they have pre-saged a mini-revolution in how type 2 diabetes is managed. After the phase 3 registration trials of twice-daily subcutaneous exenatide, the long-acting compound was studied in the phase 1 and 2 DURATION-1 (Effects of Exenatide Long-Acting Release on Glucose Control and Safety in Subjects with Type 2 Diabetes Mellitus) trial, a randomized trial consisting of a 30-week controlled phase in which patients with type 2 diabetes who had received glucose-lowering therapies (e.g., metformin or sulfonylureas) were assigned to receive either long-acting exenatide (2 mg once weekly) or short-acting exenatide (10 μ g twice daily).^{19,31} Glycated hemoglobin levels were reduced by 1.5 to 1.9%, with the greatest effect occurring with the longer-acting exenatide. Subsequent trials with liraglutide confirmed that long-acting GLP-1 receptor agonists exerted greater and more sustained glucose-lowering effects than conventional therapy and were superior to exenatide.^{32,33} The 56-week SUSTAIN 3 trial showed that 1 mg of semaglutide administered once weekly reduced glycated hemoglobin levels significantly more than 2 mg of extended-release exenatide administered once weekly.³⁴ In the PIONEER 2 trial, oral semaglutide was superior to empagliflozin, a sodium–glucose cotransporter (SGLT2) inhibitor, for reducing levels of glycated hemoglobin.³⁵ Other, larger trials provided clear support for the superiority of longer-acting GLP-1 receptor agonists over alternative second-line therapies besides metformin for type 2 diabetes.³⁶

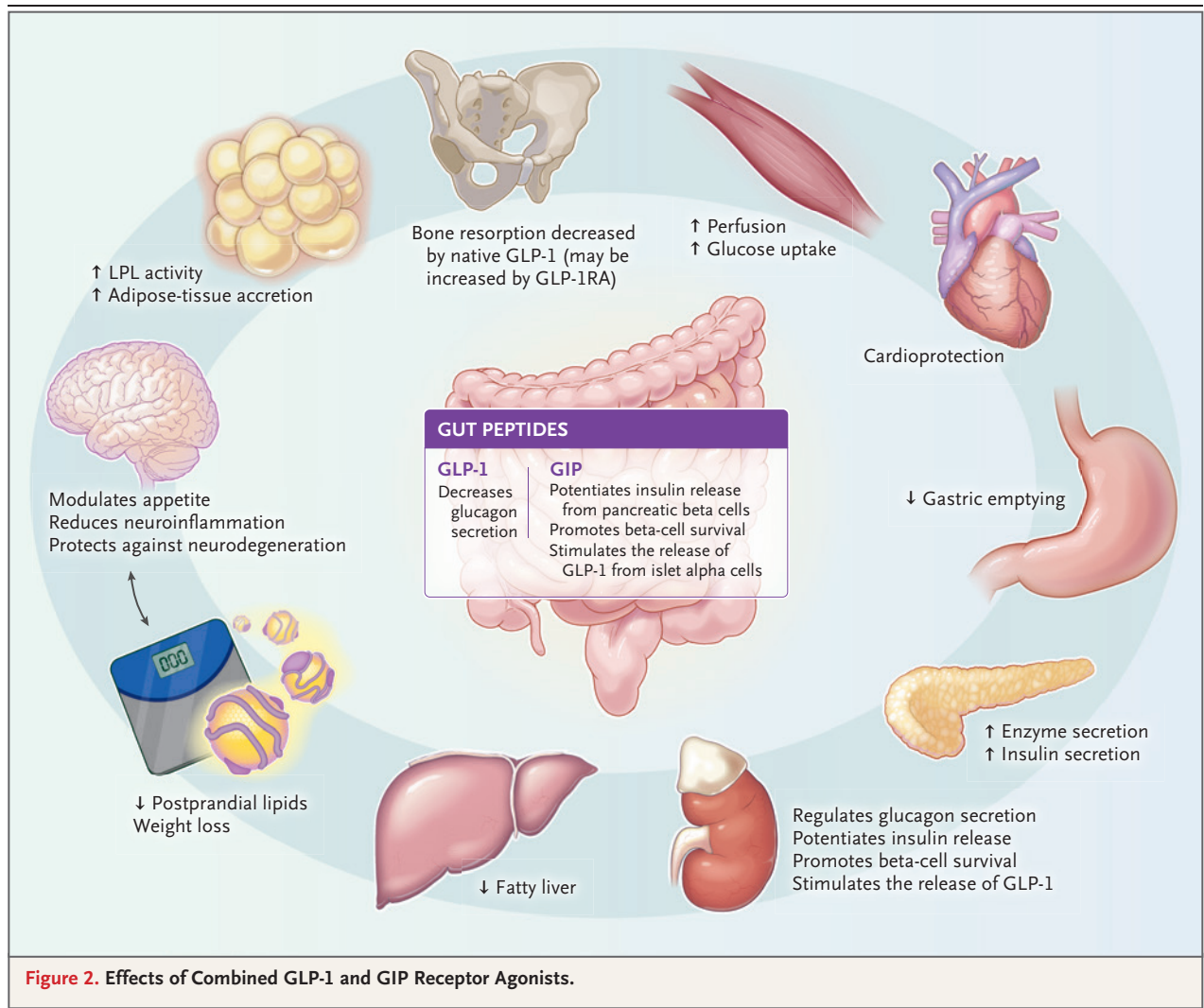
Tirzepatide was the first dual GLP-1 and GIP receptor agonist approved for the treatment of type 2 diabetes on the basis of the SURPASS-1 (Tirzepatide [LY3298176] in Participants with Type 2 Diabetes Not Controlled with Diet and Exercise Alone) trial, a randomized, placebo-controlled trial of tirzepatide (administered in weekly doses of 5 mg, 10 mg, or 15 mg) paired with metformin for treatment of type 2 diabetes (Fig. 2). Tirzepatide reduced glycated hemoglobin levels by as much as 2 percentage points and was superior to placebo at all dose levels.³⁷ At 40 weeks,

nearly 90% of participants had glycated hemoglobin levels of less than 7.0%, a change that was associated with profound weight loss. In the subsequent SURPASS-2 trial, tirzepatide at a dose of 5 mg, 10 mg, or 15 mg per week was compared with semaglutide at a dose of 1.0 mg per week. The 15-mg dose of tirzepatide was superior to semaglutide.³⁸ Decreased glycated hemoglobin levels similar to those seen in the SURPASS trials were noted in a recent meta-analysis of randomized trials of tirzepatide (15 mg weekly).³⁵ In that analysis, tirzepatide was similar to semaglutide at a dose of 2.0 mg per week in sustaining reduced blood glucose levels.³⁹ In addition, all three tirzepatide doses were superior to long-acting or ultra-long-acting insulins in controlling hyperglycemia.

More recently, retatrutide, a third-generation, triple GLP-1–GIP–glucagon receptor agonist, exerted excellent glucose-lowering effects among participants with type 2 diabetes in a large phase 2 randomized trial, superior to both placebo and dulaglutide.⁴⁰ On the heels of the development of newer double and triple agonists, several new algorithms for the use of GLP-1 receptor agonists have potential. Specifically, the application of these agents to prevent the conversion of prediabetes to diabetes, which occurs in approximately one in three affected persons over a 3-year period, may now be possible.⁴¹ Early reports suggest that GLP-1 receptor agonists may reverse impaired glucose tolerance in stage 2 type 1 diabetes mellitus (defined as the presence of diabetes-related autoantibodies and dysglycemia without symptoms).⁴²

CARDIAC AND RENAL EFFECTS

Tight control of glucose levels in patients with type 2 diabetes has been associated with reductions in major adverse cardiovascular events (MACE) and decreased progression to kidney failure.^{43,44} Therefore, it is no surprise that trials have indicated that GLP-1 receptor agonists not only reduce glycated hemoglobin levels but also positively affect cardiac and renal outcomes.⁴⁵⁻⁵⁴ In addition, it seems likely that the mechanisms responsible for these positive effects are more than just improvement in ambient glucose concentrations. For example, GLP-1 improves microvascular blood flow by increasing endothelial



function, causing vasodilatation and reducing systemic vascular resistance. Other studies suggest that GLP-1 receptor agonists can reduce blood pressure directly and indirectly through weight loss and sympathetic output in patients with type 2 diabetes.⁴⁵ The LEADER (Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results) trial showed both the noninferiority and superiority of liraglutide to placebo with regard to MACE.⁴⁶ Two major trials of semaglutide — SUSTAIN 6 (Trial to Evaluate Cardiovascular and Other Long-term Outcomes with Semaglutide in Subjects with Type 2 Diabetes) and the PIONEER 6 (Peptide Innovation for Early Diabetes Treatment) trial — enrolled patients with type 2 diabetes at high risk for ath-

erosclerotic complications.^{47,48} SUSTAIN 6 enrolled 3927 patients with type 2 diabetes, 83% of whom had preexisting cardiovascular disease, chronic kidney disease (CKD), or both. That trial showed a significant 26% reduction in the risk of MACE (a three-point composite of cardiovascular death, nonfatal myocardial infarction, or nonfatal stroke) among patients who received parenteral semaglutide once weekly. In the PIONEER 6 trial, oral semaglutide was noninferior to placebo for reducing MACE but did not reach superiority. In another trial involving persons with obesity and at high risk for cardiovascular disease, semaglutide at a dose of 2.4 mg per week was superior to placebo with regard to a composite end point of death from cardiovascular causes, nonfatal

myocardial infarction, or nonfatal stroke.⁴⁹ Most recently, in the Semaglutide Cardiovascular Outcomes Trial (SOUL), which enrolled nearly 10,000 patients with type 2 diabetes, oral semaglutide at a dose of up to 14 mg per day reduced the risk of a cardiovascular event by 14%.⁵⁰ In another trial, tirzepatide was found to reduce mortality, part of a composite end point in patients with congestive heart failure.⁵¹ Overall, the average duration of the major GLP-1 intervention trials for cardiovascular risk was approximately 3 years; therefore, studies of longer duration are still needed.

Trials embedded within the above-mentioned cardiovascular outcome studies and independent trials examined evidence that CKD progression was slowed or prevented in patients with type 2 diabetes. CKD progression was retarded by a mean of approximately 3 years; in some cases, progression was prevented. The kidney-sparing effects of liraglutide, dulaglutide, and semaglutide are both direct and indirect and appear, in part, independent of improved glucose tolerance. With respect to direct actions, improved endothelial function, beneficial changes in solute exchange in the kidney tubules, reduced activation of the renin-angiotensin system, and lower blood pressure contributed to a decreased prevalence of CKD among patients with type 2 diabetes treated with GLP-1 receptor agonists.^{52,53} In the LEADER trial, liraglutide was associated with less macroalbuminuria and lower rates of development and prevalence of diabetic kidney disease, although the continuous renal replacement and doubling of serum creatinine with liraglutide did not differ from placebo.⁵⁵ Similar results were noted with dulaglutide in an exploratory analysis of the REWIND (Researching Cardiovascular Events with a Weekly Incretin in Diabetes) trial.⁵⁴ In the SUSTAIN 6 trial, semaglutide was also associated with a 36% reduction in new or worsening nephropathy and a nearly 50% reduction in macroalbuminuria but no significant difference in the incidence of continuous renal replacement therapy.⁴⁷ Of note, in a meta-analysis of those three trials and others that enrolled more than 500 participants with type 2 diabetes, as well as the SELECT trial, which enrolled persons without diabetes and with obesity and cardiovascular risk factors, a composite kidney outcome including renal failure was reduced significantly (by 18%).⁴⁷ In January 2025, semaglutide was approved by

the Food and Drug Administration for reducing the risk of kidney disease in persons with type 2 diabetes.

GLP-1 RECEPTOR AGONISTS AND WEIGHT LOSS

In the first phase 3 trials of exenatide, modest weight loss was reported.¹⁶⁻¹⁸ Furthermore, several investigators reported that their patients who received exenatide not only had better glucose control but also were losing body weight, prompting larger trials focused on weight loss.⁴⁷ The clinical-trial results were impressive; in addition, the generation of the combination analogues increased their visibility and, in turn, advanced the concept of obesity as a disease rather than a condition. Initially, five double-blind, randomized, placebo-controlled trials of liraglutide were performed — the SCALE (Satiety and Clinical Adiposity Liraglutide Evidence) trials, which enrolled more than 5000 participants, some with type 2 diabetes and others with obesity but not diabetes.⁵⁶ The SCALE trials resulted in an average body-weight loss of approximately 5% with liraglutide. Subsequently, in 2014, liraglutide became the first GLP-1 receptor agonist approved for the management of obesity.

As with the SCALE trials, the STEP (Semaglutide Treatment Effect in People with Obesity) trials — several coordinated semaglutide trials that enrolled persons with obesity — also involved more than 5000 participants.⁵⁷ Participants were randomly assigned to receive subcutaneous semaglutide (2.4 mg once weekly) or placebo for 68 weeks. Participants in all groups received lifestyle intervention. The average reduction in body weight with semaglutide was 14.9%, as compared with 2.4% with placebo, in the STEP 1 trial.⁵⁸ On the basis of those data, semaglutide was approved for the treatment of obesity in 2021. Notably, in the STEP 4 trial, clinically significant weight regain occurred after discontinuation of semaglutide.⁵⁹ Semaglutide (2.4 mg once weekly) led to abatement of metabolic factors associated with liver disease, particularly hepatic fibrosis in patients with obesity.⁶⁰

Tirzepatide was approved as a weight-loss treatment in November 2023, with a starting dose of 2.5 mg per week and acceleration to a dose as high as 15 mg per week.⁶¹ Approval was based on

the results of the phase 3 SURMOUNT-1 trial, in which 2539 adults without diabetes were treated with tirzepatide for 72 weeks, with doses escalated every 4 weeks, up to 15 mg per week.⁶¹ Tirzepatide produced body-weight losses of 15 to 21%, as compared with 3% with placebo. Tirzepatide at the currently approved doses was also more efficacious than semaglutide for reducing body weight, with reductions ranging from 5.27 to 9.57 kg. With regard to differences between drugs and doses, tirzepatide at doses of 15 mg, 10 mg, and 5 mg showed greater efficacy than semaglutide at doses of 2.0 mg, 1.0 mg, and 0.5 mg, respectively.⁶² Tirzepatide therapy also led to clinically significant resolution of metabolic dysfunction–associated liver disease (formerly called nonalcoholic fatty liver disease), a frequent accompaniment of obesity.⁶³ Retatrutide, the previously mentioned third-generation triple GLP-1–GIP–glucagon agonist, also led to weight loss of nearly 25%, significantly more than dulaglutide at doses greater than 4 mg per week (up to 12 mg per week).⁵² In addition to effects on weight loss, treatment with retatrutide also led to clinically significant resolution of metabolic dysfunction–associated liver disease.⁶⁴

Taken together, these data show that treatment with GLP-1 receptor agonists is very effective in persons with obesity — with caveats. Some form of maintenance therapy to prevent weight regain is needed, but what therapy and for how long are unknown. In addition, the weight loss associated with the GLP-1 receptor agonists is accompanied by a reduction in lean mass that may have short-term and long-term effects on muscle and bone.

ADVERSE EVENTS

Adverse events are well-described and not uncommon with incretin analogues (Table 1). Gastrointestinal side effects are the most common and have been reported in virtually every trial of GLP-1 receptor agonists and GLP-1–GIP dual agonists during dose escalation. These side effects include nausea, diarrhea, constipation, and vomiting, which occur most often during the first 8 weeks of treatment. Because the mechanism of action of GLP-1 receptor agonists is to slow gastric and intestinal motility to reduce glucose absorption, it is not surprising that these

agents cause nausea and constipation. With regard to serious adverse events, retained gastric contents theoretically can lead to pulmonary aspiration, particularly at the time of surgery.⁶⁵ Most clinical trials of GLP-1 receptor agonists have shown a dropout rate of 4 to 8%, mainly owing to gastrointestinal side effects, but it should be noted that these trials are often a year or less in duration. In these trials, the prevalence of gallbladder disease has been greater with GLP-1 receptor agonists than with placebo. In a meta-analysis involving 76 trials and more than 100,000 participants, the risk of gallbladder and biliary disease was 1.5 times as high with GLP-1 receptor agonists as that with placebo, especially at the higher doses and longer treatment duration with GLP-1 receptor agonists.⁶⁶ Hypoglycemia is common in patients receiving multiple medications for diabetes, including GLP-1 receptor agonists, but only in the presence of other hypoglycemic medications. However, in a recently published study by the Department of Veterans Affairs, hypoglycemia was not associated with GLP-1 receptor agonists when other antidiabetes drugs were not also in use.⁶⁷

Pancreatitis has been reported anecdotally as a rare adverse event with GLP-1 receptor agonists, possibly as a result of cholecystitis; however, evidence from long-term randomized, controlled trials does not show a clinical difference between these agents and placebo. In the SURMOUNT-1 trial of tirzepatide, four cases of pancreatitis were reported (one each in the dose-level groups and the placebo group). In a review and meta-analysis of studies of liraglutide and semaglutide as compared with naltrexone–bupropion that was based on the electronic health records of 16 million persons, the number of reported cases of pancreatitis was higher with the GLP-1 receptor agonists.⁶⁸ Evidence is lacking with regard to an increased frequency of neoplasms, cancer, or death from the use of GLP-1 receptor agonists. In fact, reduction in body weight may ultimately lead to reduced risks of certain cancers associated with obesity.

Loss of lean mass is an important and consistent side effect caused by sustained weight loss, including from the use of GLP-1 receptor agonists.⁶⁹ There are few studies with data on the long-term consequences, particularly with regard to muscle function and bone. A recent

report showed a significant 1% loss of bone mineral density in the femur and spine after 52 weeks of treatment with liraglutide.⁷⁰ This decrease in bone density was probably the result of enhanced bone resorption, which has also been reported with semaglutide treatment.⁷¹ Long-term studies are needed to determine if the risk of fracture is increased.

DETERMINING THE APPROPRIATE USE OF GLP-1 RECEPTOR AGONISTS

Trials of GLP-1 receptor agonists for glucose control and weight loss have been highly successful. As such, early use of GLP-1 receptor agonists — singly, in combination, or as dual-action agents — has been suggested as a first-line approach. Certainly, with the advent of these drugs, the long-recommended initial use of metformin monotherapy in type 2 diabetes could be challenged; however, cost must be considered. Given the findings of recent phase 3 studies that provide evidence of renal and cardiovascular protection, the use first of a GLP-1 receptor agonist, a dual agonist such as tirzepatide, or a combination of a SGLT2i with a GLP-1 receptor agonist could be considered. In addition, the remarkable effectiveness of GLP-1 receptor agonists for weight loss is widely recognized. However, the questions of who should receive treatment with these agents and for how long are evolving.⁷² Indeed, the use of GLP-1 agonists has already created shortages, particularly because impressive weight reduction continues to be shown with their use.⁷³

Table 2 lists GLP-1 receptor agonists that are presently approved or are under development. Although this burgeoning of therapeutic options signals optimism that these drugs represent improved therapy for persons with obesity and type 2 diabetes, little is known of the actual long-term benefits and risk profiles of these agents. The gastrointestinal adverse events observed during clinical trials and among persons being treated lead to concern about long-term adherence to treatment and weight regain after discontinuation.⁷⁴ These issues give rise to other unanswered questions, such as whether these agents would be best used indefinitely or episodically and whether microdosing with less

drug or less frequent administration would be efficacious. Finally, questions persist about the legal and medical issues inherent in compounding these agents to enhance affordability while maintaining efficacy.

GLP-1 RECEPTOR AGONISTS AND HEALTH EQUITY

Underrepresented populations are often excluded from clinical trials for multiple reasons; thus, generalizations about the effectiveness of virtually any drug or intervention of necessity must be qualified. Such exclusion is particularly relevant for persons with type 2 diabetes and obesity, because Hispanic and Black persons make up a large proportion of the population with these diseases. Of importance, although the recruitment of members of minority groups into large trials has improved somewhat over the past decade, trial populations often do not reflect the truly affected populations. One substudy from the SURPASS-1 trial, which enrolled patients with type 2 diabetes, showed that Hispanic participants had responses similar to those of non-Hispanic White participants with regard to weight loss and lowering of glucose levels.⁷⁶ However, few other studies have analyzed the effectiveness of the GLP-1 receptor agonists in minority populations or have included enough Black or Hispanic participants in large trials to draw clinically relevant conclusions about efficacy. Complicating this concern is the cost of the newer GLP-1 receptor agonist agents. As more evidence accumulates regarding the safety and efficacy of long-term treatments, there should be price adjustments and plans for greater distribution of these drugs to underrepresented populations.

CONCLUSIONS

Taken together, clinical data on incretin-based agents now place them at the top of the therapeutic armamentarium for persons with type 2 diabetes or obesity (or both). The development of new dual and triple GLP-1 receptor agonists forecasts that other effective drugs may be approved after rigorous phase 2 and phase 3 trials.^{75,76} Strong evidence from large phase 3 clinical trials and numerous meta-analyses support

Table 2. GLP-1 Receptor Agonists and Dual or Triple Agonists Approved or under Development.*

Agent	Description
Approved to date	
Dulaglutide (Trulicity)	Long-acting agent
Exenatide (Byetta)	Short-acting agent; first such agent approved in the United States
Exenatide extended release (Bydureon)	
Liraglutide (Victoza, Saxenda)	Intermediate-acting agent
Lixisenatide (Adlyxin)	Once-daily dose
Semaglutide injection (Ozempic)	
Semaglutide tablets (Rybelsus)	For treatment of diabetes
Tirzepatide	Dual-receptor agonist; approved in 2022 for type 2 diabetes and in 2023 for weight loss
Under development	
Retatrutide	GIP, GLP-1, and GLP-1 receptor agonist
Orforglipron	Oral capsule of a chemical GLP-1 receptor agonist; phase 3 study of orforglipron will end in 2027
Danuglipron	Oral medication, a chemical GLP-1 receptor agonist, entering phase 2 studies
APHD-012	Once-daily oral medication; manufacturer describes it as a distal jejunal-release dextrose bead; phase 2 study nearing completion
ARD-101	Oral medication that works at intestinal bitter-taste receptors, where it activates appetite-suppressing hormones such as GLP-1, GLP-2, and cholecystokinin; phase 2 studies are ongoing
Amycretin	Oral medication that is a dual GLP-1 receptor and amylin agonist; once-daily dose, with subcutaneous injection being developed; phase 2 study stage in progress
Monlunabant	Oral medication, cannabinoid receptor-1 inverse agonist; once-daily dose; phase 2 study stage in progress
GSR-1290	Oral GLP-1 medication; phase 2 study stage in progress
Cagrilintide–semaglutide (Cagri-Sema)	Once-weekly injection that combines cagrilintide (amylin mimic) with semaglutide GLP-1 receptor agonist
Ecnoglutide (XW003)	GLP-1 receptor agonist; injectable, once weekly
Mazdutide (IBI362)	GLP-1 receptor agonist that also mimics glucagon; being studied for weight loss and type 2 diabetes in adults; phase 3 studies in China nearing completion
Survodutide	Once-weekly injection; activates GLP-1 and glucagon receptors; action similar to that of mazdutide
VK2735	Once-weekly injection; dual GLP-1 receptor agonist and GIP receptor agonist, similar to tirzepatide; oral version in development
Maridebart cafraglutide (MariTide)	Injectable monthly; mimics GLP-1 receptor agonist but blocks GIP receptor agonists
Bimagrumab	Monoclonal antibody against activin receptors; administered intravenously monthly

* This category of medications is changing rapidly, with anticipation that more agents will continue to be evaluated and approved over time. GIP denotes glucose-dependent insulinotropic peptide.

KEY POINTS

GLP-1 RECEPTOR AGONISTS

- Glucagon-like peptide-1 (GLP-1) receptor agonists work through enhanced glucose-mediated insulin secretion, augmented by double or triple GLP-1 receptor agonists and glucose-dependent insulinotropic peptide (GIP) receptor agonists.
- GLP-1 receptor agonists indirectly lower blood glucose levels by reducing gastric emptying, inhibiting glucagon secretion, and contributing to a beneficial intestinal microbiome.
- GLP-1 receptor agonists and GIP receptor agonists have direct effects on hypothalamic nuclei that enhance satiety and promote weight loss.
- Large-scale randomized, controlled trials have shown that GLP-1 receptor agonists reduce cardiovascular risk and progression to renal failure in persons at high risk and those with type 2 diabetes.
- Adverse events and side effects associated with GLP-1 receptor agonists are predominantly gastrointestinal but also include loss of muscle and bone mass.

the role of GLP-1 receptor agonists in stabilizing blood glucose levels in patients with type 2 diabetes as well as in effecting clinically important weight loss.^{47,77} Similarly, longer-term protective effects on cardiovascular and renal outcomes are impressive. Although reported safety signals are relatively reassuring, longer-term studies are needed, with a particular focus on weight regain and changes in muscle and bone mass. Health equity in trials and the marketplace is required to more broadly include persons who need these drugs.

Furthermore, GLP-1 receptor agonists also show promise in the targeting of other disorders, including neurodegenerative disorders, metabolic liver disease, heart failure, and cancer. The future looks promising as these GLP-1 receptor agonists enter a second half-century of study.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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